Angles in Cyclic Quadrilaterals

Exam code: 0607

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Written by: Amber
Reviewed by: Dan Finlay
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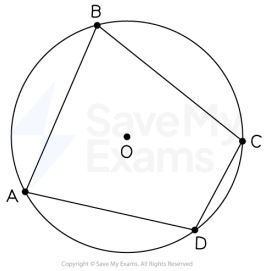
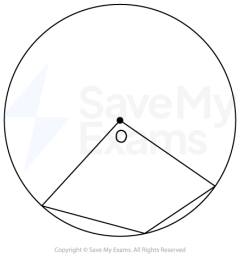
Cyclic quadrilaterals

Circle theorem: Opposite angles in a cyclic quadrilateral add up to 180°

- A quadrilateral that is formed by **four points** on the **circumference** of a circle, (a **cyclic quadrilateral**), will have pairs of **opposite angles** that add up to **180°**



- To spot this theorem in a diagram
  - look for **quadrilaterals** that have all four points on the **circumference**
- When explaining this theorem in an exam you must use the keywords:
  - **Opposite angles in a cyclic quadrilateral add up to 180°**
- The theorem only works for **cyclic quadrilaterals**
  - The diagram below shows a common scenario that is **not** a cyclic quadrilateral



Help me understand

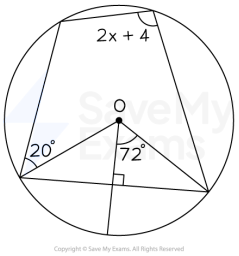
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Mark on their angles (even if you think you don't need them) as they may help you later on!

Worked Example

The circle below has centre, O.

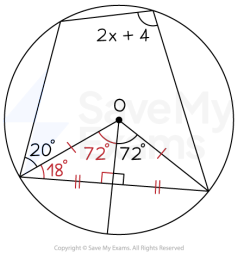
Find the value of x.



Answer:

Identify both the cyclic quadrilateral and the radius perpendicular to the chord

Add to the diagram as you work through the problem



The radius bisects the chord and so creates two congruent triangles

Use this to work out 72° (equal to the equivalent angle in the other triangle)
And 18° (angles in a triangle add up to 180°)

Then use that **opposite angles in a cyclic quadrilateral add up to 180°**

$$2x + 4 + 20 + 18 = 180$$
$$2x = 138$$

$$x = 69°$$

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